

Design Student Typologies: Beyond Perception of Generative AI

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Abstract

Generative artificial intelligence (GenAI) tools are increasingly entering illustration-oriented design education, yet uncertainty remains about how students integrate these technologies into their creative workflows. This paper examines how undergraduate design students engage with GenAI across ideation, iteration, and visualization, based on qualitative analysis of open-ended survey responses from 81 students at multiple course levels. The findings introduce two intersecting spectra that capture diverse engagement: a constructive spectrum (Starters, Accelerators, Doubters-Survivors, Finishers) and an ethical spectrum (Advocates, Guardians, Witnesses, and Suspects). These typologies illustrate how students position GenAI within both creative and ethical dimensions of practice. Rather than a uniform productivity tool, GenAI supports uses ranging from inspiration seeking and acceleration to critical, ethically reflective engagement. By foregrounding these typologies, the study offers a structured lens for understanding student-AI co-creation and provides insights for supporting reflective, responsible integration of GenAI into illustration-based studio curricula.

CCS Concepts

• Human-centered computing → Collaborative interaction.

Keywords

Generative Artificial Intelligence (GenAI), Design Education, Human-AI Collaboration, Student Typologies, Ethical Awareness

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1 Introduction and Background

In recent years, artificial intelligence (AI) tools have increasingly reshaped creative practices and design education [3, 26]. Generative AI (GenAI) technologies such as ChatGPT, Gemini, DALL-E, and Copilot are now widely used in higher education, enabling rapid idea generation through text and visual prompts [22]. Design education traditionally emphasizes creativity, originality, and critical reflection as core competencies [12, 21]. While GenAI broadens

creative possibilities, it also complicates authorship and ethical accountability [9, 26]. The opacity of generative models and their imitative nature create uncertainty around ownership, attribution, and authenticity, especially for inexperienced designers [21, 23]. Moreover, overreliance on GenAI may weaken motivation and creativity. Yet prior research has mainly focused on technical affordances rather than how students negotiate GenAI within their creative processes [5, 19, 20, 28]. Addressing this gap, this paper proposes a typology-driven approach to identify different patterns of student engagement with GenAI in illustration-based design education. Through qualitative findings, it explores how GenAI shapes creativity, ethics, and learning offering insights for its responsible integration into curricula.

Recent literature highlights four intersecting themes: GenAI's role in creative production, ethical debates around authorship and ownership, the rise of prompt literacy, and its influence on ideation and creative thinking. *GenAI in Creative Production*. GenAI enables faster, multimodal creative processes by supporting ideation, prototyping, and exploration [12, 16, 17, 25]. In creative design, it functions as both inspiration and a “creative partner,” encouraging experimentation and reflection [8, 26]. However, without critical guidance, AI-assisted ideation may lack contextual grounding, emphasizing the need for structured frameworks [28, 29]. *Ethical Considerations*. While GenAI expands creative access, it raises concerns about originality, data ethics, and authorship [1, 21]. The opacity of training data and stylistic imitation blur ownership boundaries [9], while overreliance may weaken personal style and agency [19]. Issues of data privacy and LLM accuracy further affect academic integrity [5]. Maintaining human creativity alongside AI support remains essential in illustration-based education [11, 16]. *Prompt Literacy as a Creative Skill*. Effective collaboration with GenAI depends on “prompt literacy,” or the ability to craft purposeful prompts [8, 29]. Students should learn how textual cues shape visuals; clear prompts yield relevance, poor ones generic results. Viewing AI as assistant, collaborator, or challenger aligns its use with reflective studio practice. [20, 31]. *Influence on Creative Thinking*. GenAI boosts confidence and efficiency through rapid visualization and feedback [1, 14], but risks reducing originality and intrinsic motivation [19]. Though it accelerates idea generation, GenAI's outputs may lack conceptual depth without critical interpretation [29]. Balancing automation and agency is crucial for sustaining creativity and long-term skill development [11]. The literature highlights GenAI's dual role in design education. It accelerates creative exploration but complicates authorship, ethics, and skill development. These tensions highlight the need to understand how students integrate and negotiate GenAI within studio-based learning a gap this study addresses directly.



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2 Methods

Recruitment of Participants. The survey included 81 Communication and Design undergraduates at Özyeğin University who were taking or had completed *COD 103: Illustration Fundamentals*, conducted between December 16–29, 2024, at the end of the Fall semester. The study included first-year major students (P1–P22), 1st-year minor students (P23–P36), 2nd-year students (P37–P54), 3rd-year students (P55–P67), and 4th-year students (P68–P81). Participants had previously completed illustration-oriented courses corresponding to their academic level. First-year students engaged in introductory illustration projects (COD 103), second-year students completed intermediate-level projects (COD 201–202), third-year students undertook advanced-level courses (COD 301–302), and fourth-year students participated in comprehensive studio projects (COD 401–402). This classification enabled examination of how design proficiency and creative maturity shape students' GenAI use. **Data Collection.** Data were collected through an anonymous online survey (Google Forms) available in both English and Turkish languages. Students were invited via departmental emails and class announcements. The instrument combined open-ended questions with qualitative aspects of student experiences. The open-ended section included nine questions on students' experiences, creative processes, and ethical evaluations of GenAI in illustration-based design education, organized under three themes: (1) AI tool use and experience, (2) creative thinking, and (3) ethics. Participation was voluntary, with informed consent and anonymity ensured. Responses were collected and exported to Excel for analysis. **Data Analysis.** The collected data were analyzed using qualitative thematic analysis following Byrne's six steps [4], complemented by axial coding to refine emerging typologies. All responses were first reviewed for familiarization, followed by preliminary coding of recurring ideas, expressions, and experiences. These codes were organized into potential themes reflecting patterns across academic levels. Analyses were initially conducted separately by class level to identify variations in GenAI use and perception, then merged for a holistic view of cross-level similarities and divergences overlapping categories were merged and broad ones refined for clarity. Final themes were defined and supported with representative quotations. To ensure reliability, coding was double-checked, and discrepancies resolved with a second researcher. The identified student typologies revealed connections between creativity, originality, prompt use, ethics across different patterns of student engagement.

3 Findings

The preliminary insights highlight distinct patterns in how design students engage with Generative AI during illustration-oriented design processes. The findings are structured around perceived positions that reflect variations in creative use, decision-making, and ethical awareness.

Theme 1: Integration of GenAI Across Design Phases. Students viewed GenAI as a flexible collaborator that supports exploration and refinement without replacing human authorship. *Ideation and Divergent Thinking.* In the early stages, students used GenAI as a catalyst for divergent thinking, overcoming creative blocks and expanding idea spaces. P60: "...creates things that would never come to mind and shifts my perspective." In this context, AI functioned as

a brainstorming companion rather than the originator of final ideas, raising originality concerns and prompting moderation. P24: "...good for inspiration, but it shouldn't be the final product." Accordingly, three approaches emerged: **INSPIRATION SEEKERS**, who use AI for stimulus and exploration; **BALANCED SEEKERS**, who regulate AI use to preserve agency; and **OVER-RELIANT USERS**, who depend heavily on AI outputs, risking creative erosion. *Agile Thinking and Iterative Development.* Beyond ideation, GenAI enabled iterative refinement, helping students extend concepts, identify gaps, and accelerate experimentation. P78: "...helps me notice the missing parts when I get stuck." Two iteration-oriented approaches surfaced: **AUGMENTATION SEEKERS**, who leverage AI for enhancement without surrendering control, and **DEEP DIVERS**, who iteratively refine AI outputs while maintaining creative intent. *Visualization and Implementation Support.* GenAI assisted in visualization, asset generation, and post-production. P72: "...helps me move from the sketch in my head to a more detailed visual form." AI's speed and precision were praised, although students adopted it selectively. Four strategies were identified: **SPEED SEEKERS**, optimizing efficiency; **CONCEPT POLISHERS**, refining and clarifying visuals; **PICKY USERS**, selectively integrating AI outputs; and **POST-PRODUCTION USERS**, employing AI for technical adjustments. Overall, students viewed GenAI as a catalyst for idea expansion while keeping creative judgment and final decisions human-led.

Theme 2: GenAI as a Supportive and Efficiency-Oriented Partner. While the previous section discussed GenAI's integration across design stages, this section highlights its role as a supportive, efficiency-oriented collaborator. Students viewed GenAI as assisting rather than replacing creative decision-making by reducing workload and technical barriers. *Supportive Design Partnership.* Two main approaches defined supportive use. **VISUAL SUPPORT SEEKERS** leveraged GenAI for visual ideation, producing mock-ups and references that clarified design intent. P73: "...allows me to visualize my ideas faster and see alternatives I couldn't easily draw myself." In contrast, **INFORMATIONAL SUPPORT SEEKERS** used GenAI for research and organization. P27: "...helps at the beginning, but later I need to interpret and decide on my own." Taken together, GenAI is framed as a complementary assistant supporting visualization and research, yet secondary to human interpretation. *Efficiency and Time-Saving.* Efficiency emerged as a key motivation. P7: "...makes things happen in seconds instead of hours." However, perspectives varied. **TIME-SAVERS** prioritized speed; **DUAL EVALUATORS** balanced efficiency with creative depth. P15: "It saves time, but sometimes it makes creative thinking lazy." Meanwhile, **DISTANT SKEPTICS** preferred manual control. P64: "...faster to do it with existing tools." Overall, students balanced speed, control, and creativity, revealing GenAI as both a productivity amplifier and a creative boundary marker.

Theme 3: The Originality–Creativity Dilemma. Shifting from practical to conceptual concerns, this section examines how students navigated originality and creativity in AI-supported design, as GenAI aided ideation while raising concerns about imitation and homogenization. P61: "...both support originality and dull creativity." For many, originality depended on user intention and control rather than the tool itself. *Originality and Authorship.* Two orientations emerged: **ORIGINALITY GUARDIANS**, who prioritized personal authorship and individuality P27: "...shouldn't prevent being original," and **CONDITIONAL ACCEPTORS**, who set clear boundaries

for when and how AI should be used. *Creativity Orientations*. Students' perspectives on creativity spanned a spectrum. **CREATIVITY SKEPTICS** viewed AI as constraining P80: "...it destroys creativity." **THRESHOLD PRACTITIONERS** maintained balance P48: "When used only as needed, it can have a positive effect, but constant use makes the user dependent." and **CREATIVITY OPTIMISTS** framed AI as a catalyst P19: "...can be creative, but most of the time it corrects more than it creates something new." Overall, students reframed creativity as preserving human intentionality within technological mediation.

Theme 4: Challenges in Prompt–Output Interaction. Building on earlier discussions of user control, this section examines how students experienced GenAI as a communicative partner, often struggling to articulate intent or align outputs with expectations. P79: "Not being able to explain exactly what I want or not reaching the result I hoped for." P10: "Even if I write a good prompt, it doesn't give creative ideas." In this context, prompting thus involved interpretation and trust, not mere command execution. *Prompt-Related Challenges*. Prompt formulation emerged as a key obstacle. P75: "We don't really know how to write good prompts." As a result, students reframed prompting as an expressive competence. Four approaches appeared: **ITERATIVE REFINERS**, who relied on trial-and-error; **EXPRESSION STRUGGLERS**, who struggled to verbalize visual ideas; **MISINTERPRETED USERS**, who felt misunderstood by the system; and **TRUST-DOUBTERS**, who questioned the tool's reliability. *Output-Related Challenges*. Students also noted repetitive or shallow outputs. P45: "Everyone who uses it ends up with the same style." Accordingly, three orientations emerged: **HOMOGENIZED PRODUCERS**, critical of stylistic sameness; **SURFACE CRITICS**, who acknowledged efficiency but raised concerns about superficial results; and **FAULTY-OUTPUT SURVIVORS**, who reframed AI errors as creative opportunities. Overall, prompting became a site of negotiation where creativity emerged from ambiguity in human–AI communication.

Theme 5: Ethical Boundaries Between Inspiration and Homogenization. This theme examines how students defined the ethical limits of GenAI through human contribution, ownership, and responsibility, emphasizing preserved agency and transparent use. *Creative Inspiration and Homogenization of Ideas*. Three approaches captured boundaries between inspiration and ethical risk. **CONDITIONAL-USE ADVOCATES** stressed consent and transparency. P57: "...inspiration can ...be necessary, but creating datasets from artists' works without permission or credit is not ethical." **ETHICAL GUARDIANS** accepted ideation but insisted on adding personal contribution. P23: "...getting ideas, but we ...add something of our own to distinguish it from the ready-made information it draws from." **ORIGINALITY-EROSION PESSIMISTS** warned that reliance can reduce effort and individuality. P53: "Ending the effort and individuality in design." *AI-Supported Co-Creation and Labor*. Students linked ethical boundaries to creative labor. **CO-CREATION PRAGMATISTS** viewed GenAI as ethical when acknowledged. P36: "Its contribution to the design process is ethical, as long as it's stated that the ideas come from AI." **LABOR-THREATENED** viewed substitution as unethical. P66: "Using the product of Gen AI tools as the final design and as your work is unethical." *Ownership and Attribution*. Three positions emerged: **PROMPT AND OWNERSHIP GUARDIANS** tied ownership to prompting (P68); **NEUTRAL BYSTANDERS** framed AI as a step in one's

process (P20); and **PLATFORM WITNESSES** attributed ownership to platforms or shared data (P49). *Intellectual Property and Societal Risk*. Students raised concerns about citation, consent, and misuse. **CONSENT-CREDIT ADVOCATES** foregrounded acknowledgment. P74: "Using it without citing sources... feels like a convenient workaround." **SOCIETAL-RISK WITNESSES** warned about consent violations and labor displacement. P45: "...used without artists' consent, ...companies use GenAI not as a tool but to replace designers and illustrators." Overall, students saw GenAI as a flexible collaborator that boosts ideation while challenging creativity, authorship, and ethics in design education.

4 Discussion

The findings identify eight typological positions that reflect the variation in the technical use, creativity, and ethical awareness of GenAI by students. The typologies consist of **STARTERS**, **ACCELERATORS**, **DOUBTERS–SURVIVORS**, and **FINISHERS**, as well as **ADVOCATES**, **GUARDIANS**, **WITNESSES**, and **SUSPECTS**. Synthesizing patterns across the five themes, these positions were mapped onto two intersecting spectra: a *Constructive Spectrum*, capturing how students integrate GenAI across ideation, development, and implementation, and an *Ethical Spectrum*, reflecting how they negotiate authorship, ownership, and responsibility (see Figure 1), as detailed below:



Figure 1: The Dual Spectrum of AI–Design Users: Student Typologies of GenAI Use

Ideation Phase: From Agile to Divergent Thinking. Reframing Collaboration: AI as a Design Partner in Education. Students described GenAI as a collaborative partner aiding development, presentation, and conceptual framing through rapid visuals and organization [16, 18]. Fast output enabled easier idea externalization and increased time efficiency in coursework and presentations (P23, P44). Some also used GenAI to structure projects and presentation materials (P57, P3), reinforcing its role as a supportive partner rather than a creative authority. Within the Constructive Spectrum, this pattern reflects the **STARTERS** and **ACCELERATORS** typologies: the former use GenAI for exploration and learning, while the latter emphasize efficiency and goal-oriented production. Yet students warned that such convenience may foster superficial engagement, as reflected in statements like "I did my homework" or "I passed my classes" (P38, P47). This concern aligns with the **DOUBTERS–SURVIVORS**, who adopt a cautious stance acknowledging GenAI's benefits while protecting reflection and creative depth [3, 15]. GenAI also served as a text-based assistant for writing, summarizing, and organizing research materials (P27, P20). While some likened it to "Google" (P48), others used it mainly in later stages for refinement and presentation. This selective use aligns with the **FINISHERS** typology,

who use GenAI toward the end of the process to polish content while maintaining human control [1, 3, 8, 16, 24, 27].

Negotiating Between Originality and Creativity. Students' experiences reveal a duality in how GenAI shapes originality and creativity. Some reported that AI-generated variations broadened perspectives and supported idea generation (P76, P27), while many emphasized that outputs often became repetitive and stylistically similar, leading to a perceived loss of personal originality (P45, P60, P62). Several noted that this risk depends on usage, suggesting originality can be preserved through deliberate, controlled integration (P79) [6, 9, 24]. A similar tension emerged around in relation to creativity. Many described GenAI as accelerating creative processes and supporting early ideation (P66, P39, P23), while others warned that overreliance may dull thinking and produce superficial outcomes (P33, P71). Across the Constructive Spectrum, **STARTERS** and **ACCELERATORS** emphasized inspiration and speed, **FINISHERS** regained creative control through refinement, and **DOUBTERS-SURVIVORS** resisted homogenization by foregrounding intuition. Collectively, these patterns show that students redefine creativity as directing, filtering, and evaluating AI-generated possibilities rather than mere production [13, 21, 32].

Cultivating AI Literacy through Reflective and Critical Engagement. Students' experiences indicate that effective GenAI use relies on prompt literacy, verbal expression, and reflective interaction that extend beyond technical familiarity. [2, 9, 12, 17]. Many struggled to align prompts with intended outcomes, describing misunderstandings or irrelevant results despite detailed instructions (P43, P10, P71). These challenges fostered trial-and-error learning, through which students gradually improved communication with the system, suggesting that prompt formulation itself became a learning process [9, 10, 28]. Within the Constructive Spectrum, this was most evident among **STARTERS**, who learned through experimentation, and **ACCELERATORS**, who systematized iteration for efficiency. Concerns also arose around output quality, predictability, and stylistic uniformity. Students described AI-generated visuals as repetitive or superficial (P24, P71, P58), often making them feel passive in the creative process. Many intervened manually or used post-production techniques to regain control and authorship (P26, P35). This reflective stance characterizes **DOUBTERS-SURVIVORS**, who recognize AI's limitations and turn them into opportunities for reflection and creative judgment.

Originality, Copyright, and the Redistribution of Creative Labor. Students' reflections reveal that GenAI raises ethical concerns extending beyond originality to issues of copyright, ownership, and creative labor. While AI tools were perceived as supportive for production, many questioned the fairness of systems where outputs are generated in seconds, potentially diminishing the value of human effort. As one participant stated, "It's not ethical when something people spend hours working on can be produced in seconds" (P9). This highlights that ownership involves not only authorship but also recognition of creative labor and intention [7, 22, 30]. Ethical perspectives varied across typologies. **ADVOCATES** and **GUARDIANS** emphasized human intention and originality as the basis of ownership, whereas **WITNESSES** and **SUSPECTS** perceived creative production as shared or system-oriented. Within the Constructive Spectrum, **STARTERS** and **ACCELERATORS** positioned GenAI as a functional assistant, while **FINISHERS** and

DOUBTERS-SURVIVORS foregrounded human contribution and questioned ethical boundaries.

Across these themes, students' experiences suggest that GenAI is reshaping design learning not by replacing creativity but by reframing how it is practiced and understood. The emerging typologies reflect a shift from tool adoption to reflective negotiation, aligning with HCI research on perceived ownership in co-creation and the relationship between AI intervention and human agency, where intention, authorship, and ethical awareness increasingly shape creative agency.

5 Conclusion and Future Works

The Constructive-Ethical Spectrum proposed in this study offers educators a conceptual lens for interpreting students' diverse engagements with GenAI. These patterns highlight how design learners negotiate creativity, authorship, and ethics when collaborating with generative tools. By identifying recurring typologies, it provides insight into the evolving relationship between human creativity and AI assistance. Insights from this study can inform curriculum development and teaching strategies that foster critical reflection, responsible tool use, and creative agency in design education. This study was limited to students in the Department of Communication and Design at Özyeğin University. Future research could extend to multiple institutions to examine typology patterns and uncover emerging distinctions across educational contexts. Additionally, engaging design educators in follow-up studies may reveal how these typologies can guide the integration of GenAI into curricula and studio-based practices.

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